SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 14:45 UTC

SEC P vs NP — daily log 2026-04-28

18 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) × Boolean formula depth lower bounds for KRW-style block compositions f ∘ g, the central complexity-theoretic object behind P ⊄ NC^1; equivalently the depth complexity D(f ∘ g) of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded**: 2026-04-28 01:39 UTC
- **Entry ID**: `8047f7f8eea1`

Statement

For all boolean functions f:{0,1}^n→{0,1} and g:{0,1}^m→{0,1} for which HX^1(X_f) and HX^1(X_g) admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace_pairing on the KW-metric spaces obeys the sub-additive bound κ(f ∘ g) ≥ κ(f) + κ(g) − C, where C is a universal constant independent of n,m, and where (X_{f∘g}, d_{f∘g}) is built by the coarse_product operation. In particular, witness cocycles c_f ⊠ c_g ∈ HX^1(X_{f∘g}) and tensor Roe operators T_f ⊗ T_g ∈ C_u^*[X_{f∘g}] realize log|⟨c_f ⊠ c_g, T_f ⊗ T_g⟩| / log(propagation(T_f ⊗ T_g)) ≥ κ(f) + κ(g) − C.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 (κ ≤ depth) to recover the KRW conjecture D(f ∘ g) ≥ D(f) + D(g) − O(1). The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: ⟨c_f ⊠ c_g, T_f ⊗ T_g⟩ = ⟨c_f, T_f⟩ · ⟨c_g, T_g⟩ (Connes-style product), while the propagation of T_f ⊗ T_g equals propagation(T_f) · propagation(T_g) under the depth-additive metric of coarse_product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has κ = O(1), so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse_product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
         ^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) at build_X_d before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix build_X_d to enumerate vertices correctly (e.g., X = [tuple(b) for b in itertools.product([0,1], repeat=n)]) and rerun the 25-pair × 5-seed protocol.

Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity × Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. AC^0[⊕] regime, expressed via the coarse 1-cohomology HX^1(X_f) and the Roe-trace pairing on C_u^*[X_f]
- **Recorded**: 2026-04-28 03:39 UTC
- **Entry ID**: 264c40c0d040

Statement

Let f_n = ⊕-tree_n be the balanced binary tree of XOR gates on n = 2^k inputs, with KW-metric space (X_{f_n}, d_{f_n}). Define the coboundary δ : C^0_coarse(X_{f_n}) -> C^1_coarse(X_{f_n}) and the propagation-R Roe subalgebra C_u^*[X_{f_n}]_R. Then there exists a universal constant α ∈ (0,1) such that, for all R ≤ α·log_2 n, every coarse 1-cocycle c with propagation ≤ R is a coboundary (so its class in HX^1 is trivial), while for R ≥ k = log_2 n there exist nontrivial classes c whose Roe-trace pairing satisfies log|⟨c,T⟩| / log(propagation(T)) ≥ k − O(1). Consequently κ(f_n) ≥ log_2 n − O(1), matching depth(f_n) = Θ(log n) up to constants and certifying axiom A1 sharply for an explicit recursive function.

Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known (depth(⊕-tree_n) = log_2 n).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation R ≈ log_2 n, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that κ is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of ℓ_p uniform Roe algebras

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on R*(n), dim HX^1_R, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across n ∈ {4,8,16,32}.

Multiplicity-Doubling Lower Bound from Protocol Pullback on Indexing Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Roe-Algebra methods in Communication Complexity × Deterministic two-party communication complexity CC(f∘G^n) of lifted boolean functions
- **Recorded**: 2026-04-28 08:37 UTC
- **Entry ID**: `6a6a970e1f5b`

Statement

Let G = IND_b be the standard indexing MetricGadget on X = {0,1}^b × [b], Y = [b] with d the Hamming metric on X×Y, and let f: {0,1}^n → {0,1} have deterministic decision-tree complexity Q(f). Then for the lifted input space ((X×Y)^n, d_⊕) at scale R = diam(G), every deterministic protocol Π computing f∘G^n satisfies CoarsePullbackProtocol(Π) = (m_Π, R_Π) with m_Π ≥ 2^{Q(f)} / poly(b, n), so by axiom A2, CC(f∘G^n) ≥ Q(f) − O(log(bn)). Equivalently, CLC(f, IND_b) ≥ Q(f) · log_2(asdim_R(IND_b^{⊗n})+1) − O(log bn), recovering the Raz–McKenzie bound through the coarse-cover route.

Rationale

This sub-conjecture directly tests axiom A2 (Protocol → Cover) on the canonical gadget for which Raz–McKenzie lifting is known to hold, using axiom A3 (Asdim Amplification) to guarantee that the iterated indexing gadget has unbounded coarse dimension proportional to n. If A2 is sound, the multiplicity of the protocol-induced RoeCover must scale exponentially in the query complexity, otherwise one could collapse a query-tree lower bound through a low-multiplicity cover — contradicting the structure theorem at the heart of CGL. Confirming the bound on small parameters provides empirical evidence that the coarse-cover viewpoint reproduces the standard query-to-communication lift, a necessary sanity check before pursuing axiom A4's non-relativizing strengthening.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2012.03415v1] On Query-to-Communication Lifting for Adversary Bounds

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38168056.py", line 114, in <module>
    seeds = [int(x) if x else 11 for x in input().split()]
                                          ^^^^^^^
EOFError: EOF when reading a line
```

Judge reasoning

The test crashed with an EOFError while reading seeds from stdin before any data could be produced, so neither the support nor falsification conditions can be evaluated. | next: Patch the harness to provide default seeds (or read from argv/env) when stdin is empty, then rerun the ≥20-trial battery over b∈{2,3,4}, n∈{2,3,4}.

Σ^b_0-PIND Width-2 Closure Round-Count vs DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic / Cook-Reckhow-Krajicek correspondence: the Σ^b_0-PIND round count r_2(F), defined as the minimum k ∈ ℕ ∪ {∞} such that iterating the closure operator Φ(S) = S ∪ {C : |C| ≤ 2 and C is RUP-derivable from F ∪ S in one unit-propagation pass} k times yields ⊥ ∈ Φ^k(∅); r_2 captures the depth of sharply-bounded polynomial-time

induction in PV restricted to width-2 reasoning (Cook-Reckhow lineage; rarely measured directly as a quantitative invariant on small unsatisfiable CNFs). × Lex-DPLL search-tree leaf count L(F) on small unsatisfiable 3-CNF formulas under a fixed lexicographic variable order with unit propagation, equivalently tree-resolution leaf count up to constants.
- **Recorded**: 2026-04-28 08:46 UTC
- **Entry ID**: `fc9124ecdaa8`

Statement

For every unsatisfiable 3-CNF F on $n \leq 14$ variables and $m \leq 6n$ clauses: (a) if $r\_2(F) < \infty$, then $\log_2 L(F) \leq 2 \cdot r\_2(F) + 2 \cdot \log_2(n+1)$; (b) if $r\_2(F) = \infty$, then $\log_2 L(F) \geq n/4$. Equivalently, finiteness of width-2 RUP closure forces a polynomial DPLL bound, while non-closure forces an exponential one — a sharp dichotomy.

Rationale

Width-2 RUP closure is the propositional shadow of Cook's PV restricted to the sharply-bounded Σ^b_0-PIND fragment, the weakest layer of the Buss hierarchy. The Cook-Reckhow-Krajicek correspondence predicts a polynomial relation between sharply-bounded PV-proof depth and tree-like resolution size; quantifying it should expose the exact slack and yield a dichotomy mirroring the classical PHP/Tseitin obstructions to weak proof systems.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [2407.17947v2] Supercritical Size-Width Tree-Like Resolution Trade-Offs for Graph Isomorphism - [1502.02131v2] Extracting verified decision procedures: DPLL and Resolution - [1411.7087v6] Consistency proof of a fragment of PV with substitution in bounded arithmetic - [2306.08535v1] Cyclic proofs for arithmetical inductive definitions

Empirical Test

- exit code: `124`, elapsed: `180.07s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support nor falsification conditions could be evaluated. The critic also challenges, and no instances were verified. | next: Reduce the search space (e.g., cap n at 10 and reduce seeds per (n,α) pair) or parallelize/optimize the r_2 closure and DPLL leaf-count routines so the test completes within the time budget.

Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF_min

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Number theory of polynomial heights — the (logarithmic) Mahler measure m(p) = log|a_d| + sum_{|α|>1} log|α| over the complex roots of p ∈ Z[x] (Lehmer-Boyd-Smyth tradition: Lehmer's problem on smallest non-cyclotomic measure, Boyd's heights, Smyth's resolved cubic case). Rarely deployed in meta-complexity; computable in poly time via numerical root-finding on the explicit degree-(2^n − 1) integer polynomial. × DNF_min(f): the minimum number of product terms in a DNF representation of a Boolean function f:{0,1}^n→{0,1} given its

2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy at the heart of meta-complexity worst-case-to-average-case reductions.
- **Recorded**: 2026-04-28 09:18 UTC
- **Entry ID**: `a6aec390c941`

Statement

Define the truth-table polynomial p_f(x) := sum_{i=0}$^{\{2}$n − 1} f(i)·x^i ∈ Z[x], where f(i) is the value of f on the n-bit binary expansion of i. For every f with |f^{-1}(1)| ≥ 1, m(p_f) ≤ (DNF_min(f) − 1) · log_2(n + 1). Equivalently, every f whose truth-table polynomial has logarithmic Mahler measure exceeding (k − 1)·log_2(n+1) satisfies DNF_min(f) > k.

Rationale

Each DNF product term corresponds to a coordinate-aligned subcube whose index set is a coset a+W ≤ F_2^n; its indicator polynomial factors as x^a · prod_j (1 + x$^{\{2}${i_j}}), so all roots lie on the unit circle and m = 0 (purely cyclotomic). Disjunction of k subcubes makes p_f a signed sum of k such cyclotomic-rooted polynomials, and Boyd-style sub-additive heuristics for Mahler measure under integer-coefficient sums predict at most a log_2(deg+1) escape per added term. If true, this transforms a hard Diophantine height into a poly-time-computable lower bound on DNF_min, giving a Lehmer-type obstruction useful in restricted-MCSP regimes.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Plünnecke-Ruzsa Doubling of Constraint Vectors Lower-Bounds SoS Degree for 3-XOR

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics in F_2^n: the Plünnecke-Ruzsa doubling constant K(A)=|A+A|/|A| and its log version σ(A)=log_2|A+A|-log_2|A| of a finite set A⊆F_2^n (Freiman-Ruzsa / Bogolyubov / Sanders’ polynomial Bogolyubov tradition). The mathematical branch is harmonic / arithmetic combinatorics on abelian groups, distinct from polyhedral / Newton-polytope geometry, and rarely deployed in SoS lower bounds for CSPs. × Sum-of-Squares refutation degree d_SoS(F) of unsatisfiable 3-XOR instances F over GF(2), accessed via the Grigoriev-Schoenebeck combinatorial proxy g*(F) = min{|T| : ⊕*{i∈T} a_i = 0 ∈ F_2^n and* ⊕{i∈T} b_i = 1 ∈ F_2}, which sandwiches SoS degree to within ±1 by the Schoenebeck (2008) characterization for 3-XOR.
- **Recorded**: 2026-04-28 09:45 UTC
- **Entry ID**: `6ae7590ec4aa`

Statement

For every unsatisfiable 3-XOR instance F = ((a_i,b_i))_{i=1..m} on n ≤ 20 variables with constraint multiset A = {a_1,...,a_m} ⊆ F_2^n, the Schoenebeck combinatorial SoS-degree satisfies g*(F) ≥ σ(A) + 1, where σ(A) = log_2|A+A| - log_2|set(A)|. Equivalently, |A+A| ≤ |set(A)|·2^{g*(F)-1}. A single (F, A) violating this inequality refutes the conjecture.

Rationale

Schoenebeck's theorem says small SoS degree forces a short additive dependency among constraint vectors; Plünnecke-Ruzsa says short additive dependencies push A toward an affine-subspace structure with low doubling. The bridge predicts a monotone, never-before-quantified link between structurelessness (large doubling) of the constraint vectors and the unavoidable length of a Schoenebeck certificate, transferring Freiman-Ruzsa intuition into proof complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2507.01005v1] Discovery and study of the eclipsing variable star Grigoriev 1 - [0701009v2] Electron mobility on a surface of dielectric media: influence of surface level atoms - [1812.10216v2] Robust Approach for Rotor Mapping in Cardiac Tissue - [1701.04521v1] Sum of squares lower bounds for refuting any CSP - [0305179v3] Polynomial Degree and Lower Bounds in Quantum Complexity: Collision and Element Distinctness with Small Range

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 70, in run_trial
    set_A_size, A_plus_A_size, sigma_value = sigma(A)
                                             ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 56, in sigma
    A_plus_A.add(xor(a, b))
                 ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 21, in xor
    return x ^ y
           ~~^~~
TypeError: unsupported operand type(s) for ^: 'list' and 'list'
```

Judge reasoning

The test crashed with a TypeError in the xor() helper (xoring two lists), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the xor() helper to operate on integer bitmask representations of F_2^n vectors (or convert lists via tuple/int) and re-run the 3000-instance sweep.

Kolmogorov Width of Lifted Functions Bounds MA^cc

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Descriptive complexity theory via Kolmogorov complexity × MA communication complexity
- **Recorded**: 2026-04-28 10:22 UTC
- **Entry ID**: `dd203ee122c9`

Statement

For any Boolean function f, the Kolmogorov width of the lifted function f ∘ G^n is at least Ω(log(n)) times the MA communication complexity of f, where G^n is the n-bit Indexing gadget.

Rationale

Kolmogorov complexity and descriptive complexity theory provide a framework for understanding the intrinsic complexity of objects, which can be applied to communication complexity to derive lower bounds. The use of lifting gadgets, such as the Indexing gadget, allows us to amplify the complexity of the function, making it more amenable to analysis.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1811.04010v2] The layer complexity of Arthur-Merlin-like communication - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [0506068v1] Quantum Arthur-Merlin Games - [1409.0584v2] Kolmogorov structure functions for automatic complexity - [1101.2946v3] Quantum Kolmogorov Complexity and Information-Disturbance Theorem

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 43, in run_trial
    lifted_f = [f[(i >> n) * (1 << n) + i % (1 << n)] for i in range(2**(n+n))]
                ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support nor falsification conditions could be evaluated. No empirical evidence exists to assess the conjecture. | next: Fix the indexing bug in the lifted function construction (the formula `(i >> n) * (1 << n) + i % (1 << n)` is malformed for the Indexing gadget) and rerun the 120 trials.

Tensor-Amplified Asdim Forces Linear Communication Blow-up on Inner-Product Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Asymptotic Dimension (geometric group theory applied to communication complexity) × Deterministic two-party communication complexity of lifted functions f∘G^n, with G obtained by k-fold GadgetComposition of a base inner-product gadget IP_2

- **Recorded**: 2026-04-28 10:39 UTC
- **Entry ID**: `916caa07c776`

Statement

Let IP_2 be the 2-bit inner-product MetricGadget on X=Y={0,1}^2 with Hamming metric d, and let G_k = IP_2^{⊗k} under coordinatewise additive metric. For every Boolean f:{0,1}^n→{0,1} with deterministic query complexity Q(f), the CoarseLiftingComplexity satisfies CLC(f, G_k) ≥ Q(f) · log_2(k+1) − O(n), and consequently CC(f∘G_k^n) ≥ Q(f)·log_2(k+1) − O(n). Equivalently, AsdimCertify(LiftedInputSpace(G_k), R=diam(G_k)) returns multiplicity m ≥ k+1, so by axiom A2 every deterministic protocol Π must have cost ≥ log_2(k+1) per query bit recovered.

Rationale

This sub-conjecture directly tests axiom A3 (Asdim Amplification) chained with A2 (Protocol→Cover). A3 predicts that iteratively tensoring IP_2 (which already has asdim ≥ 1 because its 16-point input space contains a metric line of length 4) yields G_k with asdim ≥ k − O(1). A2 then converts the cover multiplicity into a communication lower bound. If the conjecture holds, it provides a concrete constructive route to lifting bounds that scale with the tensor depth k, recovering Raz–McKenzie–style log-factor blow-ups from a purely coarse-geometric witness rather than from gadget-specific combinatorics. Failure would falsify A3 in the small-multiplicity regime and force a refinement of how asdim composes under ⊗.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [1104.4468v3] A strong direct product theorem for quantum query complexity - [2511.17227v3] A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [1110.6876v4] Lower bounds for topological complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 124, in run_trial
    G_k = gadget_composition([ip_2] * k)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 74, in gadget_compos
    result[i*n+k][j*n+l] = gadgets[i][k] + gadgets[j][l]
                           ~~~~~~~~~~^^^
TypeError: 'function' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError in gadget_composition (treating a function as subscriptable) before any data was produced, so neither the support nor falsification criteria could be evaluated. | next: Fix the gadget_composition implementation to operate on metric matrices/arrays rather than function objects, then re-run the 12 (k,n,f) instances across 3 seeds.

Möbius Defect of NW Design Lattice Bounds Parity Bias

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Möbius theory of finite lattices (Rota–Stanley combinatorial Möbius inversion on intersection lattices; Stanley EC1 Ch.3) — rarely deployed in PRG bias analysis × Output bias of the Nisan–Wigderson pseudorandom generator NW_D$^{f:\{0,1\}}$d→{0,1}^m against linear (parity) distinguishers χ_T, T⊆[m]
- **Recorded**: 2026-04-28 10:48 UTC
- **Entry ID**: `b27f028c321c`

Statement

Let D=(S_1,…,S_m) be an NW design over universe [d] with |S_i|=n and pairwise intersections ≤k. Let L(D) be the meet-semilattice of all distinct intersections T_I=□*{i∈I} S_i (I⊆[m]) ordered by ⊆ with bottom 0̂=□, and let μ_L be its Möbius function. Define the Möbius defect δ(D)=Σ*{T∈L(D), T≠□} |μ_L(0̂,T)|·2^{−|T|}. Conjecture: for every f:{0,1}^n→{0,1} with max Walsh coefficient |f̂(U)|≤2^{−n/3} on |U|≥k, and every non-empty T⊆[m], |E_{y~U_d}[χ_T(NW_D^f(y))]| ≤ 4·δ(D).

Rationale

Standard NW bias bounds use the crude factor m·2·ε from a hybrid argument that ignores higher-order inclusion–exclusion cancellation across the design's intersection pattern. Möbius values μ_L(0̂,T) are precisely the signed multiplicities by which intersections of design rows are over/under-counted, so replacing m by δ(D) should yield substantially tighter bias estimates whenever L(D) is structured (e.g., projective-plane or polynomial designs versus random subset-systems). If the conjecture holds, δ(D) becomes a computable, design-only certificate of fooling power independent of f beyond the Fourier-tail hypothesis.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2601.03730v1] Perception-Aware Bias Detection for Query Suggestions - [1207.0393v2] Pseudo-finite hard instances for a student-teacher game with a Nisan-Wigderson generator - [2109.14047v1] Second Order WinoBias (SoWinoBias) Test Set for Latent Gender Bias Detection in Coreference Resolution - [0710.4339v1] Heavy-Quark Masses from the Fermilab Method in Three-Flavor Lattice QCD - [1111.0981v1] Form factors for B to Kll semileptonic decay from three-flavor lattice QCD

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
```


Judge reasoning

The test stdout contains only `{"TRIAL": null}` entries with no measurements, no RESULT line, and no evidence that the pre-registered criteria (≥200 designs, both constructions, regression slope ≥0.5, zero violations) were evaluated. The critic also flags a likely vacuous-bound saturation issue, so SUPPORTED cannot be claimed. | next: Fix the test harness so trials actually emit (D,f,T,bias,δ(D)) tuples, then verify whether 4·δ(D) is non-trivial (<1) for the tested regime before re-running the p

Sandpile-Group Order of Certificate Conflict Graph Lower-Bounds Q^dt

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Abelian sandpile models / chip-firing on finite graphs (Bjorner–Lovász–Shor; Holroyd–Levine–Mészáros–Peres–Propp–Wilson); the sandpile (critical) group K(G) computed as the cokernel of the reduced graph Laplacian, equivalently the spanning-tree count by Kirchhoff's matrix-tree theorem × Decision-tree depth Q^dt(f) of small Boolean functions f:{0,1}^k→{0,1}, transported via the Raz–McKenzie / Göös–Pitassi–Watson Indexing-gadget lifting theorem to deterministic two-party communication complexity D^cc(f∘IND_b)
- **Recorded**: 2026-04-28 11:19 UTC
- **Entry ID**: `ef886a7094f0`

Statement

For every Boolean f:{0,1}^k→{0,1}, let G_f be the graph whose vertices are the minimal 1-certificates of f and whose edges {C,C'} connect certificates that fix some common variable to opposite values (call these 'conflicts'). Then log_2 |K(G_f)| ≤ Q^dt(f) · log_2(k+1), where |K(G_f)| is the order of the abelian sandpile group of G_f, which equals the number of spanning trees of G_f by Kirchhoff (defined to be 1 if G_f has ≤1 vertex or is disconnected with each component a tree). By GPW lifting, this entails D^cc(f∘IND_b) ≥ Ω(log_2 |K(G_f)| · log b / log k).

Rationale

The minimal 1-certificate clutter captures the irreducible branching obligations any decision tree must witness; their conflict graph encodes exactly the pairs that cannot coexist on a single root-to-leaf path, so its cycle-richness should constrain protocol/decision-tree branching budgets. The sandpile group is a torsion abelian invariant whose log-size is Kirchhoff's spanning-tree count — a single integer determinant — and a depth-d tree induces at most k^d distinct query histories, plausibly bounding how much such cyclic redundancy can be 'covered'. Chip-firing and sandpile invariants have essentially never been deployed in lifting-style lower bounds, giving an algebraic-combinatorial obstruction distinct from rank, Dilworth-width, Möbius, or asymptotic-dimension routes.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0906.2809v2] Sandpile groups and spanning trees of directed line graphs - [2602.01042v2] On Condensation of Block Sensitivity, Certificate Complexity and the `AND` (and `OR`) Decision Tree Co - [1904.12209v2] Sandpile monomorphisms and limits - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [0811.2800v2] Parallel Chip-Firing on the Complete Graph: Devil's Staircase and Poincare Rotation Number

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 131, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 104, in run_trial
    if not is_minimal_1_certificate(f, C):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 25, in is_minimal_1_
    if f[C[:i] + (1 - C[i],) + C[i+1:]] == f[C]:
          ~~~~~~^~~~~~~~~~~~~
TypeError: can only concatenate list (not "tuple") to list
```

Judge reasoning

The test crashed with a TypeError (list/tuple concatenation bug in is_minimal_1_certificate) before any of the 800 trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the type bug by representing certificates consistently (e.g., always tuples) so that C[:i] + (1 - C[i],) + C[i+1:] type-checks, then rerun the 800-trial sweep.

Stanley-Reisner Projective Dimension Lower-Bounds Monotone-KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial commutative algebra: projective dimension pd(R_Δ) of the Stanley-Reisner ring R_Δ = GF(2)[x_1,…,x_n]/I_Δ accessed via Hochster's formula β_{i,T}(R_Δ) = dim tilde H_{|T|-i-1}(Δ|_T; GF(2)), in the Stanley–Hochster–Reisner–Terai tradition (Miller–Sturmfels, Combinatorial Commutative Algebra). Rarely deployed against circuit/formula complexity; distinct from polytope/Newton-polytope or matroid-rank approaches. × Monotone Karchmer–Wigderson game depth D_m(f) of monotone Boolean functions f:{0,1}^n→{0,1}, equivalently monotone formula depth in the {AND,OR}-basis via the Karchmer–Wigderson equivalence — the central object of Razborov-style monotone NC^1 lower bounds.
- **Recorded**: 2026-04-28 11:47 UTC
- **Entry ID**: `e180d6571db0`

Statement

Define Δ_f := {S ⊆ [n] : f(1_S)=0}, the simplicial complex of non-1-certificates of a monotone f, and let Pd(f) := max{|T|−j−1 : T ⊆ [n], j ≥ −1, tilde H_j(Δ_f|_T; GF(2)) ≠ 0} (with Pd(f)=0 if all reduced homologies vanish). Then for every monotone f, the monotone-KW protocol depth satisfies D_m(f) ≥ Pd(f). One counterexample (a monotone f with D_m(f) < Pd(f)) refutes the conjecture.

Rationale

A monotone-KW protocol decomposes the rectangle MIN(f)×MAX(¬f) into coordinate-labeled monochromatic rectangles; this rectangular decomposition is the combinatorial shadow of a free resolution of I_{Δ_f}, whose length is exactly pd(R_{Δ_f}) by Auslander–Buchsbaum. Hochster's formula reduces pd to vanishing of reduced GF(2)-homology on induced subcomplexes, an invariant that is rarely linked to KW games (prior monotone bounds use fooling sets, lattice rank, or Razborov approximators). Because the conjecture targets monotone classes only, it sidesteps the natural-proofs barrier (which constrains P/poly with arbitrary basis under crypto assumptions, not monotone NC^1).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1101.4831v1] The *f*–vector of the clique complex of chordal graphs and Betti numbers of edge ideals of uniform hypergraphs

Empirical Test

- exit code: 1, elapsed: 0.11s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 75, in run_trial
    D_m_f = compute_D_m(f, n)
            ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 22, in compute_D_m
    min_f = [f(i) for i in range(2**n)]
             ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 19, in <lambda>
    return lambda s: all(x in s for x in range(n))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_850c8ff1.py", line 19, in <genexpr>
    return lambda s: all(x in s for x in range(n))
                         ^^^^^^
TypeError: argument of type 'int' is not iterable
```

Judge reasoning

The test crashed with a TypeError in the f-construction lambda before any instance was evaluated, so no data was produced and the pre-registered support condition was not met. | next: Fix the buggy monotone-f constructor (the lambda treats integer states as iterable) and rerun the full 20+168+500 enumeration.

Lie-Stabilizer Dimension of Clause Cubic Bounds DPLL Leaves

- **Verdict**: INCONCLUSIVE
- **Bridge**: Geometric Complexity Theory: dimension of the stabilizer Lie algebra g_p = {A ∈ Mat_n(Q) : Σ_i (A·x)_i ∂_i p(x) ≡ 0 as a polynomial} of a homogeneous polynomial p, computed by Gaussian elimination on the linear system extracted from the cubic-monomial coefficients; the canonical first-order Mulmuley–Sohoni / Bürgisser–Ikenmeyer obstruction used to separate det_n from perm_n via stabilizer dimension ($2n^2-1$ vs. ~2n−2). Distinct from apolar/catalecticant rank and from Hessian rank. × Lex-DPLL search-tree leaf count L_DPLL(F) on small UNSAT 3-CNF formulas at clause density α, equivalently tree-resolution leaf count up to constants — the canonical proof-complexity proxy for refutation hardness.
- **Recorded**: 2026-04-28 12:19 UTC
- **Entry ID**: 77253411c65d

Statement

Define p_F ∈ Q[x_1,...,x_n] by p_F(x) = Σ_{clauses C} ε_C · x_{i_1}x_{i_2}x_{i_3}, where ε_C = (−1)^{#negated literals in C} and (i_1,i_2,i_3) are the indices of C's variables; let g_F be its GL_n stabilizer Lie algebra and dim_Q(g_F) its dimension. Conjecture: for every UNSAT 3-CNF F on n ≥ 8 variables, log_2 L_DPLL(F) + dim_Q(g_F) ≤ n + 2·⌈log_2 n⌉. A single UNSAT F violating this inequality (with positive margin) refutes it.

Rationale

GCT identifies high stabilizer dimension with maximally structured polynomials (the determinant's huge symmetry is exactly what trivializes its computation), and the orbit-closure containment program reads lower bounds off symmetry-budget mismatches. Translated to refutations, an UNSAT formula whose clause cubic carries large Lie symmetry should expose unit-propagation/decomposition structure, while symmetry-poor cubics correspond to the 'rigid' regime where DPLL is forced to branch combinatorially. Cubic clause polynomials are the cheapest GCT-style invariant capturing 3-SAT, and their stabilizer Lie algebras have not, to my knowledge, been linked to proof complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1401.3795v2] Relationship between Nichols braided Lie algebras and Nichols algebras - [1611.00827v2] Geometric complexity theory and matrix powering - [0911.3482v5] Complexity of Networks (reprise)

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92d0288f.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92d0288f.py", line 74, in run_trial
    M = [[0] * (n**2) for _ in range(binomial(n+2, 3))]
                               ^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

The test crashed with a TypeError (binomial returned a float passed to range) before any trials were run, so no Δ(F) values were produced and neither the support nor falsification condition can be evaluated. | next: Fix the binomial call by casting to int (or use math.comb) and rerun the 1000-instance sweep to obtain actual Δ(F) statistics.

Følner-Defect Floor Forces Logarithmic Communication Overhead on Equality-Lifted Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — coarse geometry / geometric group theory applied to communication complexity × Deterministic two-party communication complexity CC(EQ_k ∘ G^n) of the lifted Equality function on n copies of a fixed MetricGadget G
- **Recorded**: 2026-04-28 12:42 UTC
- **Entry ID**: `723804a64233`

Statement

Let G = (X,Y,g,d) be a MetricGadget with |X|=|Y|=k and Hamming-graph metric d, and let EQ_k : {0,1}^n → {0,1} denote Equality on n bits with Q(EQ_k) = n. Define the Følner defect of the lifted space at scale R as δ_R(n) = min over finite F ⊂ (X×Y)^n of |∂_R F|/|F|, where *∂R F is the d*⊕-R-boundary. Then for every gadget G satisfying axiom A4 (Property A) and every n, CC(EQ_k ∘ G^n) ≥ Q(EQ_k) · ⌈log_2(1/δ_R(n) + 1)⌉ for R = diam(G), and equivalently CLC(EQ_k, G) is bounded below by n · log_2(AsdimCertify(LiftedInputSpace_n, diam(G))+1). In particular, when AsdimCertify returns multiplicity m at scale R, every deterministic protocol must use at least n · log_2(m+1) bits.

Rationale

This sub-conjecture directly tests Axiom A2 (Protocol → Cover) combined with Axiom A4 (Property-A Transfer). A2 says any cost-c protocol yields a RoeCover of multiplicity ≤ 2^c, so log_2 of the minimum cover multiplicity lower-bounds CC. A4 promotes that bound by the query complexity factor Q(f). For Equality, Q(EQ_k) = n and any non-trivial RoeCover at scale diam(G) of the n-fold lift must have multiplicity at least k^Ω(1) per coordinate by amenability obstructions (Følner defect), giving a per-coordinate logarithmic multiplier. The conjecture should hold because the Følner-defect quantity δ_R(n) shrinks geometrically in n for product metrics, forcing the cover multiplicity — and hence the protocol cost — to scale linearly with n with a logarithmic-in-k coefficient.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data - [0901.0512v4] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 123, in run_trial
    cc_eqk_gn_value = cc_eqk_gn(G, n)
                      ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d88ab7be.py", line 88, in cc_eqk_gn
    C = [[0 for _ in range(m)] for _ in range(m)]
                                        ^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

Judge reasoning

The test crashed with a TypeError before any data was produced ('float' object cannot be interpreted as an integer in cc_eqk_gn at line 88), so no configurations were evaluated and the pre-registered support criterion cannot be assessed. | next: Cast m to int (e.g., m = int(round(m))) when constructing the cost matrix in cc_eqk_gn, then rerun the 12 configurations to obtain CC vs n·log_2(m_n+1) data.

Bipartite Token-Sliding Diameter Lower-Bounds Monotone-KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial reconfiguration / token-sliding graphs (Ito-Demaine-Hearn-Etsuko-Uno tradition): the diameter D_TS(B_f) of the token-sliding reconfiguration graph on the bipartite minterm/maxterm incidence structure of a monotone Boolean function f, where tokens sit on minterms (1-side) or maxterms (0-side) and a slide moves a token along a 'shared coordinate' edge. × Monotone Karchmer-Wigderson game depth D_m(f) for monotone Boolean functions, equivalently monotone formula depth in the {AND,OR} basis
- **Recorded**: 2026-04-28 12:48 UTC

- **Entry ID**: `299d5162a4f6`

Statement

For every monotone Boolean function f on n variables, the monotone-KW depth satisfies D_m(f) >= log2(D_TS(B_f) + 1) - 1, where D_TS(B_f) is the diameter (in the connected component containing the canonical all-coordinates token configuration) of the token-sliding graph on the bipartite minterm-maxterm incidence B_f under the move 'slide token from minterm m to minterm m' iff m and m' differ in exactly one coordinate i with f(m')=1, and similarly on the maxterm side'. Equivalently, no monotone formula of depth d can resolve every (minterm, maxterm) pair within fewer than 2^(d+1)-1 token slides, so depth d < log2(D_TS+1)-1 is forbidden for every f. One reconfiguration-diameter witness D_TS(B_f) > 2^(d+1) refutes any claimed monotone formula of depth <= d for f.

Rationale

A monotone-KW protocol of depth d defines a labeling of (minterm, maxterm) pairs by coordinates that respects f-monotonicity; concatenating coordinate-labels along any reconfiguration path yields a protocol-consistent path on the incidence bipartite graph, and protocol depth bounds the number of distinct decisions reachable by any single token-slide trajectory. Reconfiguration diameters are well-studied combinatorially but, to our knowledge, have never been pulled back through the KW correspondence as a depth lower-bound certificate, despite providing a direct combinatorial obstruction (long sliding paths force deep protocols). The bridge isolates a non-spectral, non-monotone-circuit-counting invariant that is robust under taking sub/superfunctions and sidesteps the natural-proofs barrier because D_TS is not closed under random monotone projections.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2504.10671v2] Token Sliding Reconfiguration on DAGs - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1904.06696v3] On bipartite distance-regular Cayley graphs with diameter 3 - [2302.03695v4] The Harer-Zagier and Jackson formulas and new results for one-face bipartite maps

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
{"TRIAL": {"seed": 11, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 23, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 37, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 53, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
{"TRIAL": {"seed": 71, "metric_name": "monotone-KW depth", "metric_value": null, "instances_tested":
```


Judge reasoning

All 5 trials returned metric_value=null, instances_tested=0, and counterexample='mapping_undefined', meaning the test never computed D_m(f) or D_TS(B_f) for any function. No empirical evidence was produced, and the pre-registered threshold of >=25 functions was not met. | next: Disambiguate the token-sliding move rule (especially the maxterm-side clause) and implement a concrete D_TS computation plus a monotone-KW depth oracle for small n (3..5) before scaling to n=8.

Magnus Level-2 Defect Bounds DNF_min via Truth-Table Inversions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Magnus expansion / Chen's iterated-integral signature in noncommutative formal power series of free groups (Magnus 1935; Chen 1957; Reutenauer, Free Lie Algebras 1993; Lyons rough-path tradition). The level-2 'inversion defect' $\lambda_2(w) = |\#\{i<j : w_i=0, w_j=1\} - \#\{i<j : w_i=1, w_j=0\}|$ is exactly the pairing of the truncated Magnus signature $S^{\le 2}(w) \in \mathbb{Q}\langle\langle A,B\rangle\rangle$ with the commutator $[A,B]$, a free-Lie-algebra invariant computable in one $O(|w|)$ pass and rarely deployed in meta-complexity. × DNF_min(f), the minimum number of product terms in a DNF representation of a Boolean function $f:\{0,1\}^n \to \{0,1\}$ presented by its 2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy used in meta-complexity worst-case-to-average-case reductions and the canonical truth-table-input meta-complexity object.
- **Recorded**: 2026-04-28 13:27 UTC
- **Entry ID**: `8bf8c5c64dba`

Statement

Let $T = TT(f) \in \{0,1\}^{2^n}$ be the lex-ordered truth table of f and define $\lambda_2(f) = |\mathrm{asc}(T) - \mathrm{desc}(T)|$ where $\mathrm{asc}(T) = \#\{i<j : T_i=0, T_j=1\}$ and $\mathrm{desc}(T) = \#\{i<j : T_i=1, T_j=0\}$. Then for every Boolean f on n variables, $\log_2(1 + \lambda_2(f)) \le n + \log_2(1 + \mathrm{DNF\_min}(f)) + 5\cdot\log_2(n+1)$. A single f violating this inequality refutes the conjecture; for uniformly random f one has $\log_2 \lambda_2(f) \approx 1.5\, n$ by anti-concentration, so the bound is exponentially below random and forces $\mathrm{DNF\_min}(f) = 2^{\Omega(n)}$ only on the trivial counting fraction.

Rationale

The Magnus expansion linearizes a binary word into a free Lie algebra; its level-2 $[A,B]$-coefficient is the inversion-pair imbalance of the truth-table string, computable in a single $O(2^n)$ sweep. Low-DNF functions have truth tables decomposing as the union of M structured sub-rectangles whose individual inversion imbalances align in a near-cancelling way, so λ_2 should grow like $M\cdot 2^n$ rather than the random $2^{1.5n}$. The hardness predicate '$\lambda_2(f) \ge 2^{n+\log M+O(\log n)}$' is satisfied by a $1-\exp(-\Omega(n))$ fraction of random f only as a trivial counting bound, and the small-λ_2 side (which is the structural side actually used to upper-bound DNF_min) holds for an exponentially tiny fraction of random f, dodging the Razborov–Rudich largeness condition and hence the natural-proofs barrier.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0810.5488v1] The Magnus expansion and some of its applications - [1501.01331v1] DNF complexity of complete boolean functions - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [1509.08963v3] Integrals of Lipschitz-Hankel Type, Legendre Functions, and Table Errata - [2402.02935v4] Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even-Z nuclei

Empirical Test

- exit code: `0`, elapsed: `146.77s`


```
{"TRIAL": {"seed": 11, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.0276395303248895, "ins
{"TRIAL": {"seed": 23, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.00788077049084, "insta
{"TRIAL": {"seed": 37, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.00585013217493, "insta
{"TRIAL": {"seed": 53, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.041945775829029, "inst
{"TRIAL": {"seed": 71, "metric_name": "lambda_2 vs DNF_min", "metric_value": 7.017263172398402, "inst
RESULT: SUPPORTED mean=7.020115876243618 std=0.013375328575307719 support_fraction=1.0
```

Judge reasoning

The test reports SUPPORTED with only 16 instances per seed at unspecified n, falling far short of the pre-registered ≥840 instances across n∈{8,…,20} and four ensembles; the critic's saturation argument (log_2(1+λ_2) ≤ 2n makes the bound vacuous until n≥28) means the tested regime cannot probe potential failures. The pre-registered support condition is not unambiguously met. | next: Run the full pre-registered protocol at n∈{8,10,12,14,16,18,20} with ≥30 seeds per (n,ensemble,M) cell, explicit

---

Monomer-Dimer Entropy Lower-Bounds Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Statistical mechanics on graphs / monomer-dimer theory: Heilmann-Lieb real-rootedness of the matching polynomial m_G(x)=Σ_k (-1)^k m_k(G) x^{n−2k} and the resulting per-site free energy h_md(G)=(1/n) log Σ_k m_k(G), a Heilmann-Lieb / Godsil / Gurvits invariant rarely deployed in proof complexity and distinct from spectral expansion. × Resolution refutation width w(T(G,c)⊢⊥) of Tseitin formulas on 3-regular graphs G with odd total F_2-charge c — the Urquhart / Ben-Sasson–Wigderson / Alekhnovich–Razborov target.
- **Recorded**: 2026-04-28 13:54 UTC
- **Entry ID**: `c5c4b106a0cf`

Statement

There exists an absolute constant α > 0 such that for every 3-regular graph G on n vertices and every charge c ∈ F_2^{V(G)} with Σ_v c_v = 1, the resolution refutation width satisfies w(T(G,c)⊢⊥) ≥ α · n · h_md(G), where h_md(G) := (1/n) · log_2 Σ_{k=0}^{⌊n/2⌋} m_k(G). A single 3-regular graph on at most 20 vertices with odd charge whose Tseitin formula admits a resolution refutation of width strictly below 0.05 · n · h_md(G) refutes the conjecture.

Rationale

Heilmann-Lieb log-concavity makes h_md a topological-combinatorial 'routing entropy' of unit F_2-flows in the cycle space, and these routings are exactly the obstruction narrow resolution must clear to cancel an odd parity. Unlike edge expansion, h_md is girth-sensitive and survives moderate bottlenecks, so the conjecture predicts width bounds strictly stronger than Ben-Sasson–Wigderson on low-expansion-but-high-h_md graphs such as long prisms or generalized Möbius-Kantor families. The tie to extended Frege is downstream: if the bound saturates the BSW expansion bound only on Ramanujan-type graphs, the slack pinpoints where higher proof systems can compress.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [2309.13632v2] Decay of correlations in the monomer-dimer model

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 189, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 159, in run_trial
    h_md_value = h_md(G)
                 ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_31fe51cd.py", line 52, in h_md
    u, v = e
    ^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

The test crashed with a ValueError in the h_md function (edge unpacking failure) before producing any data, so no instances were evaluated against the pre-registered criterion. | next: Fix the edge iteration in h_md (handle multi-edges/self-loops or unpack edges correctly) and rerun the experiment over the specified 3-regular graphs and odd charges.

Sprague-Grundy Game Value of CNF Bounds Tree-Frege Lines

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sprague-Grundy / Nim-value theory of impartial combinatorial games (Bouton 1901; Sprague 1935; Grundy 1939; Berlekamp-Conway-Guy 'Winning Ways' tradition); a branch of combinatorial game theory rarely deployed in proof complexity, distinct from Karchmer-Wigderson games and from Ehrenfeucht-Fraisse pebble games used in descriptive complexity. × Tree-like Frege refutation line count L_TF(F) on small unsatisfiable 3-CNFs, accessed via Reckhow's polynomial equivalence with tree-resolution leaf count L_T(F⊢⊥) (depth-0 Frege over CNFs), serving as a tractable proxy for the Krajicek-Buss-Razborov-Impagliazzo EF lower-bound program.
- **Recorded**: 2026-04-28 14:25 UTC
- **Entry ID**: `501b089a21b9`

Statement

Define the impartial Variable-Splitting Game V(F) on a CNF F as follows: a state is a CNF obtained from F by a sequence of literal substitutions with clause simplification (drop satisfied clauses, drop falsified literals from surviving clauses); a move from state F' picks any variable v occurring in F' together with a value b∈{0,1} and transitions to F'[v=b]; terminal states (no remaining variables) carry Sprague-Grundy value 0, and non-terminal values are the mex of children's values. Let G(F) denote this Grundy value. Conjecture: for every unsatisfiable 3-CNF F with n ≤ 20 variables, ⌊log_2(1+G(F))⌋ ≤ ⌈log_2 L_T(F⊢⊥)⌉, where L_T is the leaf count of lex-DPLL with unit propagation; a single unsatisfiable 3-CNF violating this inequality refutes the conjecture.

Rationale

Sprague-Grundy values capture an exponential structural complexity intrinsic to the move-graph of CNF splitting, yet have never been linked to proof-system line counts; if the inequality holds, G(F) yields a Frege-style certificate of refutation hardness computable without exhibiting any refutation, opening a non-Boolean-Fourier route from impartial-game algebra to EF size lower bounds via Reckhow's reduction. The Grundy value is invariant under literal-flipping and variable-renaming, matching a symmetry class compatible with natural-proofs-safe invariants. Failure modes (small G(F) with large L_T) are equally informative, exposing a genuine gap between game-theoretic disjunctive complexity and proof-theoretic refutation length.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1912.12573v3] A base-p Sprague-Grundy type theorem for p-calm subtraction games: Welter's game and representations of generalized symm - [1702.07068v1] The Sprague-Grundy function for some nearly disjunctive sums of Nim and Silver Dollar games - [1507.04665v2] Invariance of the Sprague-Grundy Function for Variants of Wythoff's Game - [2207.14140v1] Playing a 2D Game Indefinitely using NEAT and Reinforcement Learning - [1104.3098v3] Optimal strategies for a game on amenable semigroups

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 107, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 83, in run_trial
    F = generate_cnf(n, alpha)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3b71d3.py", line 66, in generate_cnf
    v1, v2 = random.sample(variables, 2)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError in generate_cnf (random.sample called on a non-sequence) before any instances were evaluated, so neither the violation count nor the Pearson correlation was measured. | next: Fix generate_cnf to pass a list (e.g., random.sample(list(variables), 2)) and rerun the pre-registered 300-instance sweep.

Roe-Skeleton Spectral Gap Forces Multiplicity in Protocol-Induced Covers on Small-Diameter Gadget Lifts

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe-Algebra Operator Theory applied to Lifting in Communication Complexity × Deterministic two-party communication complexity of f∘G^n, specifically the multiplicity profile m_Π of the ProtocolPullback partition on the LiftedInputSpace (X×Y)^n
- **Recorded**: 2026-04-28 14:45 UTC
- **Entry ID**: a510fd856726

Statement

Let G = (X, Y, g, d) be a MetricGadget with R := diam(G) and let A_R denote the 0/1 adjacency matrix of the RoeAlgebraSkeleton at scale R on (X×Y)^n under the d_⊕ metric, with combinatorial Laplacian L_R = D_R − A_R. Let λ_R(n) be the smallest nonzero eigenvalue of L_R, and let N = |X×Y|^n. Then for every deterministic communication protocol Π computing any non-constant f∘G^n, the ProtocolPullback multiplicity satisfies m_Π · R_Π ≥ ⌈ λ_R(n) · N / (4 · max-degree(A_R)) ⌉, and consequently the CoarseLiftingComplexity invariant obeys CLC(f, G) ≥ log_2(λ_R(n) · N /

$(4\cdot deg))$ for every f with $D(f) \geq 1$. Equivalently, a vanishing Roe-skeleton spectral gap is a coarse obstruction that lower-bounds protocol cover multiplicity, in alignment with axiom A2 (Protocol → Cover) and A4 (Property-A Transfer).

Rationale

This concretizes axiom A2 by predicting a quantitative spectral certificate: the Roe algebra skeleton's Cheeger-type gap controls the minimum multiplicity any transcript-induced cover must exhibit at scale R. The Cheeger inequality on the skeleton graph implies the boundary-to-bulk ratio of any low-diameter set is at least $\lambda_R/(2\cdot deg)$; since a transcript pullback of scale R_Π and multiplicity m_Π yields a cover where each cell has $d_\oplus$-diameter $\leq R_\Pi$, the spectral gap forces the cover cells to be 'fat' — i.e., to overlap. A protocol with multiplicity 1 (a partition) at scale R is precluded whenever $\lambda_R > 0$, certifying that asdim ≥ 1 in the lifted geometry. This also gives a lattice on which Property-A (axiom A4) can be cross-checked, since amenable/Property-A spaces have shrinking gaps but persistent multiplicity, providing a concrete bridge between coarse spectral data and communication lower bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1903.06084v8] Lifting coarse homotopies - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ce6686.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ce6686.py", line 119, in run_trial
    for gadget, X, Y, d, R in [(X_XOR, Y_XOR, d_XOR, R_XOR), (X_IND, Y_IND, d_IND, R_IND)]:
        ^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 5, got 4)
```

Judge reasoning

The test crashed with a ValueError (tuple unpacking mismatch) before producing any data, so neither the support nor the falsification condition can be evaluated. | next: Fix the iterable in run_trial to include the gadget label (5-tuple) and rerun the deterministic test to obtain $\lambda_R(n)$, N, and max-degree values.